Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Linear Independence
- Understanding Linear Independence in Vector Spaces — Reading • 18 min
- Linear Independence, Basis, and Rank — Ungraded App Item • 20 min
- Linear Independence — Reading • 30 min

Linear Mapping
- Exploring Transformations: Understanding Linear Mappings — Reading • 1h
- Linear Mapping: Part 1 — Video • 8 min
- Linear Mapping: Part 2 — Video • 8 min
- Linear Mapping — Reading • 5 min
- Matrix Representation — Reading • 6 min
- Practice Quiz: Linear Mapping — Graded Assignment • Grade: 100%

Lesson 3: Eigenvalues and Eigenvectors
- Introduction to Eigenvalues and Eigenvectors — Reading • 40 min
- Eigenvectors and Eigenvalues: Examples and Applications — Reading • 30 min

Lesson 4: Cholesky Decomposition
- Cholesky Decomposition — Reading • 5 min
- Solved Examples — Reading • 10 min

Lesson 5: Singular Value Decomposition
- Introduction to Singular Value Decomposition (SVD) — Reading • 28 min
- Derivation of Singular Value Decomposition (SVD) from Eigenvalues and Eigenvectors — Reading • 15 min

Module 3
Probability Foundations for Statistical Learning

Lesson 1: Sum Rule, Product Rule, and Bayes' Theorem
- Sum Rule, Product Rule, and Bayes' Theorem — Reading • 15 min
- Sum Rule — Reading • 4 min
- Product Rule (Chain Rule) — Reading • 2 min
- Bayes' Theorem — Reading • 43 min

Lesson 2: Gaussian Distribution
- **Univariate Gaussian Distribution** — Reading • 4 min

Univariate Gaussian Distribution

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The univariate Gaussian distribution describes the distribution of a single random variable X with two parameters: the mean μ and the variance σ^2.

- The distribution is symmetric around the mean μ.The mean μ is the location parameter, and the variance σ^2 controls the spread.
- Approximately 68% of the data falls within one standard deviation ($\mu \pm \sigma$) of the mean, 95% within two, and 99.7% within three (this is known as the 68-95-99.7 rule).

Probability Density Function (PDF)

$$\mathrm{P}\left(\mathrm{x} \mid \mu, \sigma^2\right) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\mathrm{x}-\mu)^2}{2\sigma^2}\right)$$

Where:

- x is the variable.
- μ is the mean.
- σ is the standard deviation.

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